

CHING-WEI (WILLIAM) KANG

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EDUCATION

University of Wisconsin-Madison

B.S. Computer Science and Data Science | GPA: 3.75/4.0

Relevant coursework: Data Structures & Algorithms, Artificial Intelligence, Computer Engineering, Discrete Math, Statistical Modeling, Linear Algebra.

Madison, WI

Expected May 2027

ENGINEERING INTERESTS

Focus areas: AI systems and agent orchestration; backend and data infrastructure; LLM observability and local-model tooling; computer vision and 3D reconstruction; brain-computer interfaces; reinforcement learning; trustworthy, human-reviewable software.

SKILLS

Programming: Python, TypeScript, Java, SQL, Go, C++

Technologies: Next.js, React, Node.js, FastAPI, Express, PostgreSQL, Redis, SQLite, Docker, AWS, REST APIs, RBAC, SSE, CI/CD

OPEN SOURCE ENGINEERING

- **Merged NVIDIA/NemoClaw PR #6084**, adding package CLI alias shims to improve installer command availability across supported invocation paths.
- **Merged Docker CLI PR #6957**, updating image-push auxiliary notes to respect NO_COLOR behavior and maintain consistent terminal output.
- **Merged Microsoft APM PR #1993**, adding a fast path for universal applyTo matching to improve compile-path behavior in a production developer tool.
- **Merged GitHub Docs PR #43969**, clarifying when private actions can be referenced directly so developers can apply repository-access behavior correctly.
- **Merged Alumnium PR #395**, adding Selenium proxy and user-agent options to an MCP-driven browser automation framework.
- **Merged Cloudflare vinext PR #2521**, propagating resolved Next.js server externals into Nitro trace dependencies while preserving and de-duplicating user configuration; added regression coverage across default packages, custom externals, development mode, and production builds.
- **Merged Microsoft APM PR #1971**, enforcing least-privilege executable permissions and normalizing deny-list identities to prevent bypasses through casing, .git suffixes, and alternate GitHub URL forms.
- **Merged NVIDIA NemoClaw PR #5676**, hardening agent-list parsing against warning text and multiple JSON response schemas while preserving agent-ID filtering; validated with targeted tests, linting, CLI builds, and type checking.
- **Merged Meta DotSlash PR #16**, adding portable SHA-256 verification for downloaded release assets with sha256sum and shasum support across macOS, Windows, and Linux release targets.
- **Contributed through maintainer review, automated tests, and CI workflows** across GitHub Docs, Meta DotSlash, Repomix, NVIDIA tooling, and other production repositories.

PROFESSIONAL EXPERIENCE

Full-Stack Engineering Intern | Global AI (Global API Inc.)

New York, NY | Hybrid | May 2026 - Present

- **Worked directly with the CEO on an internal pilot producing ~50 product videos/day**, each about 20 seconds; expanded one image into 4–8 keyframes and cut generation time from 20+ to ~12 minutes for faster product-demo iteration.
- **Cut per-video model cost by 70–80%** through specialized agent stages, cached Gemini outputs, reusable intermediate assets, and model routing that avoided repeated high-cost calls while preserving output quality across repeated generation runs.
- **Raised internal-pilot job completion from 90% on first attempt to 99% after retries**, recovering 90% of failed jobs through stage-level checkpoints, idempotent retries, and resumable execution while limiting unrecoverable failures to 1%.
- **Enabled faster, higher-quality team delivery** by packaging the workflow into a reusable Ad Studio with per-stage reruns and visible intermediate outputs; product demos generated collaboration interest from 2–3 startup prospects during early customer outreach.

AI & Software Engineering Intern | Guangdong Akang Health Technology Group Co., Ltd. China | On-site | Jul 2026 - Aug 2026

- **Supported a multi-source research data platform** that ingested 1M+ web documents from 30+ feeds and standardized 4.5M+ pharmaceutical, legal, company, and partner records into reusable JSON and SQLite datasets for internal analysis.
- **Maintained review-ready company intelligence** across 150K businesses and ~450GB of stored data, supporting ~50K new records/day with source IDs, URLs, retrieval dates, validation metadata, and reproducible exports for research teams.
- **Built multilingual translation and model fine-tuning workflows for a WeChat Mini Program**, helping employees review and reuse pharmaceutical and company research with more consistent domain terminology using human review checkpoints.
- **Presented AI fundamentals to 100+ managers** and separately briefed senior leadership on export-policy data collection, analysis, and model fine-tuning progress, connecting technical findings and policy implications with overseas expansion planning.

SELECTED PROJECTS

Multi-Agent OpenClaw

Jan 2026 - Present

Technologies: TypeScript, Node.js, Express, Zod, REST APIs, persistent memory, automated tests

- **Built a 20-endpoint orchestration system** that classifies tasks by type and complexity, assigns specialized agent roles, and executes configurable research, synthesis, review, and QA stages.
- **Added schema validation**, persistent JSONL run memory, context retrieval, dynamic model selection, QA-gated Markdown/DOCX exports, and seven regression tests to make agent workflows inspectable and repeatable.

PromptTrace - LLM Observability

2026

Technologies: TypeScript, Node.js, JSONL tracing, latency/cost analytics, local dashboards

- **Built a local-first tracing SDK and dashboard** that records model calls, retries, P50/P95 latency, endpoint-level cost estimates, and failure categories so developers can debug reliability and spending.

- **Protected developer data** through prompt redaction, SHA-256 hashing, bounded local logs, configurable retention, and JSON/Markdown exports without requiring a cloud observability account.

Badminton AI Match Analysis

2026

Technologies: Python, OpenCV, YOLO11, tracking, pose estimation, 3D reconstruction

- **Developed stable two-player tracking, custom shuttle detection, pose overlays, and annotated video outputs** for repeatable match review and model debugging.
- **Implemented homography-based court calibration**, single-view 2D-to-3D skeleton lifting, labeled-frame benchmarks, and GitHub Actions regression gates validated against court geometry.

Brain-Computer Interface MVP

2026

Technologies: Python, FastAPI, Streamlit, EEG/EDF, scikit-learn, Docker

- **Built an EEG pipeline** for EDF preprocessing, relaxed-versus-focused classification, real-time debounced inference, and interactive neural-signal visualization.
- **Evaluated cross-subject generalization** with leave-one-subject-out workflows and implemented RF/SVM/MLP stacking, CORAL domain adaptation, SHAP analysis, tests, and reproducible release checks.
- **Published a Dockerized FastAPI and Streamlit demo** with model-card, limitations, quality-scorecard, and release-readiness documentation to make experimental results easier to inspect and reproduce.

ClearMove - Rental Evidence Platform

Oct 2025 - Present

Technologies: Next.js, TypeScript, PostgreSQL, Drizzle ORM, Gemini, pdf-lib

- **Shipped a deployed, mobile-first evidence workflow** that transforms inspection checklists, timestamped photos, notes, and lease inputs into structured, auditable PDF reports.
- **Designed 21 API routes and 10 PostgreSQL tables** for token-based onboarding, property and unit records, submissions, and document metadata, informed by 50+ stakeholder interviews.
- **Integrated Gemini to parse photographed checklists and summarize lease clauses** while preserving user review, evidence metadata, and downloadable reports for renters and property managers.

llm-fit - Local Model Hardware Advisor

2026

Technologies: JavaScript, local LLM sizing, hardware-fit estimation, Ollama/llama.cpp

- **Converted RAM, VRAM, CPU, and platform constraints into model-fit and speed estimates** with copy-paste run commands, helping users evaluate local deployment before downloading a model.
- **Designed both CLI and web workflows** around common Mac and PC hardware profiles, emphasizing practical local-AI deployment decisions over server-only benchmark results.

Reinforcement Learning for Super Mario Bros.

2026

Technologies: Python, DreamerV3, PPO, evaluation tooling

- **Benchmarked DreamerV3 and PPO** on Super Mario Bros. 1-1 and documented training behavior, reproducible evaluation results, engineering tradeoffs, and failure modes.
- **Built checkpoint rendering, evaluation, runtime diagnostics, smoke tests, and monitoring utilities** so training progress and failures could be reviewed outside the training loop.

TableUs - Group Restaurant Discovery

2026

Technologies: Next.js, React, Python, FastAPI, Gemini API, Google Maps API

- **Built an explainable, location-aware recommendation workflow** combining group preferences, geospatial context, and natural-language input.
- **Turned individual preferences into a shared group-decision flow** and earned Second Place at the 2026 Cursor Hackathon through the team-built product and demonstration.

Dungeons & Dragons Voice Adventure

2026

Technologies: TypeScript, Gemini Live, real-time voice, multimodal generation

- **Built a real-time voice-driven storytelling system** with turn-based state, live agent responses, and dynamically generated visual feedback.
- **Explored low-latency multimodal interaction and failure handling** across speech, narrative state, and generated media to keep gameplay responsive and coherent.

ClawWork Personal Agent Workspace

2026

Technologies: Python, OpenClaw, agent dashboards, evaluation workflows

- **Built a personal agent workspace** that combines OpenClaw integrations, workflow dashboards, and evaluation utilities for experimenting with repeatable AI-assisted work.

ACHIEVEMENTS

- Third Prize, 2025 Badger Build Fest Venture Track - ClearMove
- Second Place (Skills) and Third Place (Qualification), 2023 VEX Robotics World Championship